Software Engineering

Introduction

session1

Software engineering

- The economies of ALL developed nations are dependent on software.
- More and more systems are software controlled
- Software engineering is concerned with theories, methods and tools for professional software development.
- Expenditure on software represents a significant fraction of GNP in all developed countries.

Software costs

- Software costs often dominate computer system costs. The costs of software on a PC are often greater than the hardware cost.
- Software costs more to maintain than it does to develop. For systems with a long life, maintenance costs may be several times development costs.
- Software engineering is concerned with cost-effective software development.

What is software?

- Computer programs and associated documentation such as requirements, design models and user manuals.
- Software products may be developed for a particular customer or may be developed for a general market.
- Software products may be
  - Generic - developed to be sold to a range of different customers e.g. PC software such as Excel or Word.
  - Bespoke (custom) - developed for a single customer according to their specification.
- New software can be created by developing new programs, configuring generic software systems or reusing existing software.

What is software engineering?

- Software engineering is an engineering discipline that is concerned with all aspects of software production.
- Software engineers should adopt a systematic and organised approach to their work and use appropriate tools and techniques depending on the problem to be solved, the development constraints and the resources available.

What is a software process?

- A set of activities whose goal is the development or evolution of software.
- Generic activities in all software processes are:
  - Specification - what the system should do and its development constraints
  - Development - production of the software system
  - Validation - checking that the software is what the customer wants
  - Evolution - changing the software in response to changing demands.

What & Why UML?

- UML - Unified Modeling Language
- Standard notation to help designers
- **Visualize, Specify, Guide and Document complex software projects**
- UML
  - is programming language independent
  - is a combination of several diagramming methods
  - was designed by The Three Amigos:
    James Rumbaugh, Ivar Jacobson, Grady Booch.

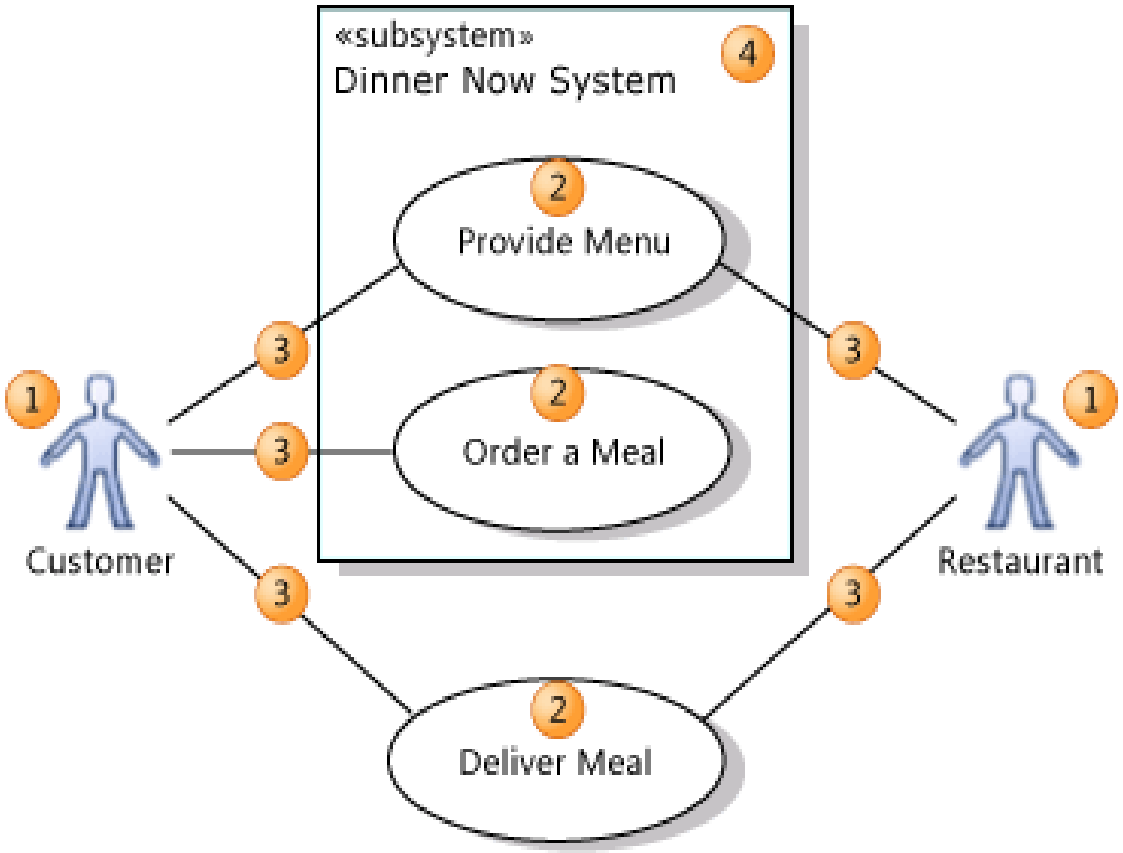
Motivations for UML

- We need a modeling language to:
  - help develop efficient, effective and correct designs, particularly Object Oriented designs.
  - communicate clearly with project stakeholders (concerned parties: developers, customer, etc).
  - Give us the “big picture” view of the project.

Types of UML diagrams

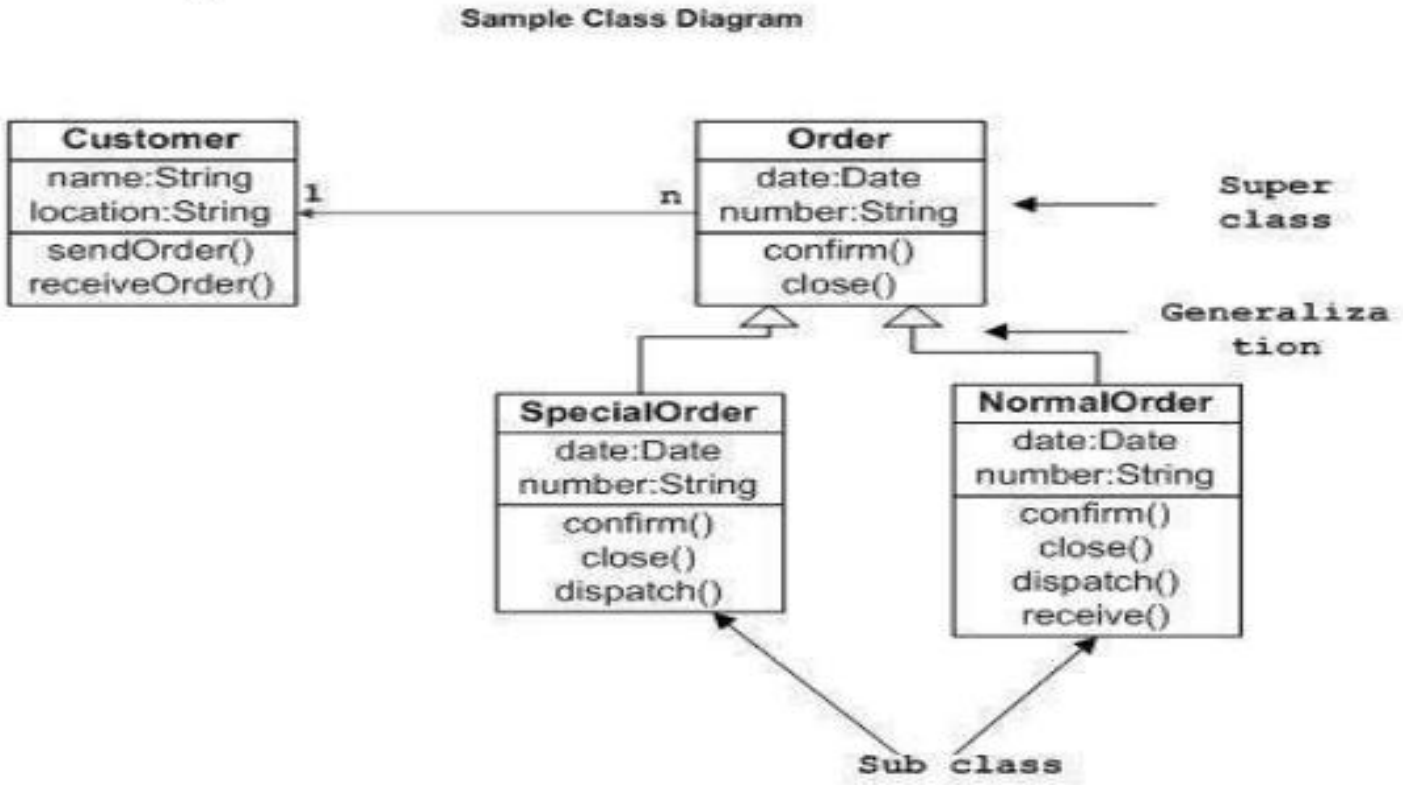
- There are different types of UML diagram, each with slightly different syntax rules:
  - use cases.
  - class diagrams.
  - sequence diagrams.
  - package diagrams.
  - state diagrams
  - activity diagrams
  - deployment diagrams.

Use case diagram

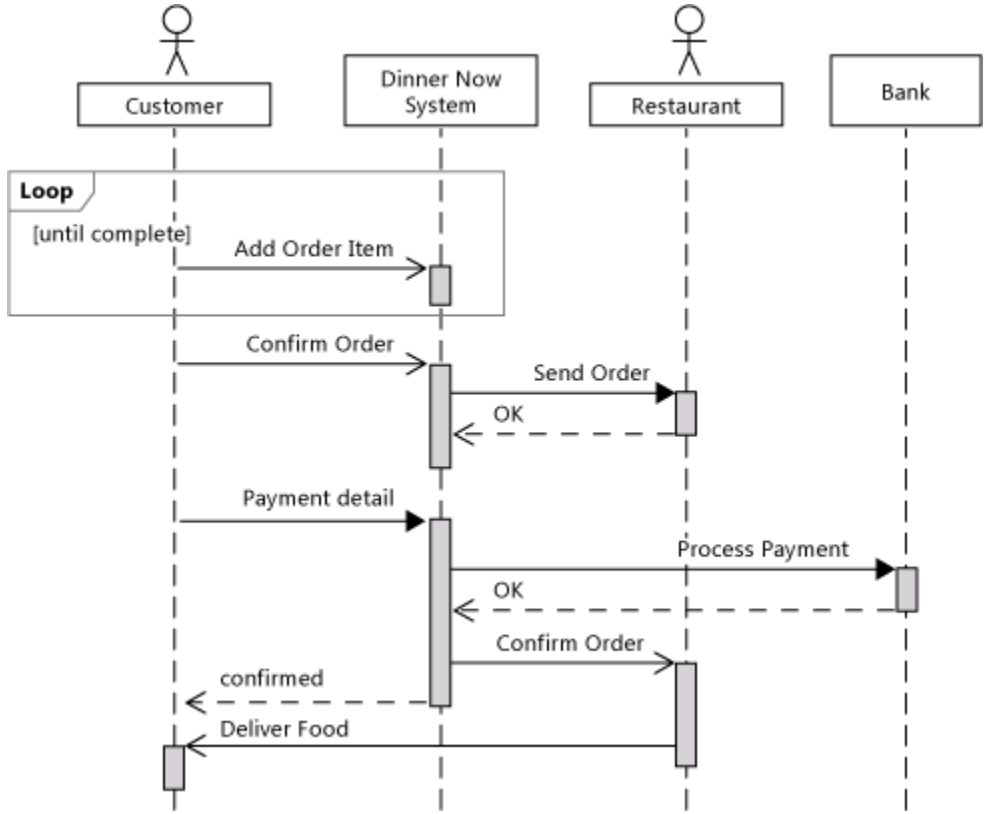


Class diagram

Example



Sequence diagram



Activity diagram

